

SnapLearn Edge

Privacy-first on-device AI lecture & meeting copilot for Snapdragon-powered HP PCs

SnapLearn Edge is a proposed privacy-first AI application for students and knowledge workers. It turns lecture, meeting and voice-note audio into a transcript, structured notes, action items, revision questions and contextual Q&A.; The product is designed around an offline-first architecture for Snapdragon-powered HP PCs so that sensitive audio and derived knowledge can remain on the device whenever possible.

Challenge Area	How SnapLearn Edge addresses it
Technical Implementation	Modular audio → ASR → transcript → local LLM → notes/Q&A pipeline with separate model-adapter layers.
Application Use Case & Innovation	Local AI knowledge companion for lectures and meetings, prioritising privacy, connectivity independence and low latency.
Deployment & Accessibility	Simple Streamlit UI, Windows/Python setup, external model folders and a documented Snapdragon integration path.
Presentation & Documentation	README, architecture, model plan, deployment guide, challenge mapping, project description and pitch deck.

Problem

Cloud-first transcription and generative-AI workflows can require users to upload sensitive recordings, depend on reliable connectivity and pay recurring inference costs. In classrooms and business meetings, those constraints can reduce trust and practical usefulness.

Proposed solution

The application provides one private workflow: capture or upload audio, transcribe it locally, structure the transcript into concise notes, identify action items, generate revision questions and answer questions grounded in the session context.

AI models and Snapdragon optimisation

The proposed speech layer uses a Qualcomm AI Hub Whisper model such as Whisper-Small-Quantized or Whisper-Base. The local reasoning layer is designed around Llama 3.2 3B Instruct. On Snapdragon X-class Windows PCs, the implementation path targets Qualcomm-compatible acceleration/runtime options and quantised models to reduce memory and power cost.

User experience

A simple desktop-friendly interface lets users upload audio, generate notes and ask questions without exposing model complexity. Outputs are intended to be exportable for revision, documentation and follow-up.

Deployment approach

The repository keeps third-party model binaries outside source control. Evaluators can install the Python dependencies, place/export compatible model assets into the documented model folders, connect the runtime adapters and launch the Streamlit interface.

Current prototype status

The repository includes a functional application UI and model-adapter scaffold. It does not claim measured Snapdragon latency, throughput or power figures. Those values should be added only after hardware-specific NPU inference is integrated and tested on the target HP Snapdragon PC.

Why it is a strong challenge fit

SnapLearn Edge directly combines an everyday productivity problem with an edge-AI architecture. The concept is practical, easy to demonstrate, privacy-oriented and intentionally structured around Snapdragon-powered HP PCs and Qualcomm AI Hub/open-source models.

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